

ANALOG ELECTRONICS

Tomorrow's Exam — Quick Revision & Most Important Questions

■ TOP PRIORITY

Focus on these first. They are the strongest recurring areas from the PYQs and syllabus reviewed.

1	CE BJT Amplifier	Midband gain + lower 3-dB frequency numerical
2	Frequency Response	Low / mid / high-frequency regions, cutoff frequencies, bandwidth
3	MOSFET Amplifier	High-frequency model, upper 3-dB frequency, gain
4	Current Mirrors	Basic BJT current mirror, current relationship, applications
5	Widlar Current Source	Circuit, working and design/derivation
6	Miller Effect	Concept, Miller capacitance and effect on high-frequency response
7	Multistage Amplifiers	CE-CE / CS-CS analysis and overall gain
8	BJT Small-Signal Model	Hybrid- π model and transition frequency

1. Frequency Response of BJT Amplifier

Prepare:

- Draw the frequency-response curve of a BJT amplifier.
- Explain the low-frequency, mid-frequency and high-frequency regions.
- Define lower cutoff frequency (f_L), upper cutoff frequency (f_H) and bandwidth.
- Know the factors affecting low- and high-frequency response.

Must remember: Bandwidth $\approx f_H - f_L$.

2. CE BJT Amplifier — Numerical

- Calculate midband voltage gain.
- Calculate lower 3-dB frequency.
- Use the small-signal / hybrid- π model.
- Identify the role of coupling and bypass capacitors.

Exam priority: VERY HIGH — practice at least 2 complete numericals.

3. MOSFET Amplifier — High Frequency

- Draw the high-frequency model of a MOSFET amplifier.
- Calculate midband gain or upper 3-dB frequency.
- Understand the effect of parasitic capacitances on high-frequency response.

4. Current Mirror / Widlar Current Source

- Define a current mirror and explain its operation.

- Derive the relationship between reference current and output current.
- Draw and explain the Widlar current source.
- Know the advantages/applications of current-source circuits.

5. Miller Effect

- Explain the Miller effect in a transistor amplifier.
- Understand Miller capacitance and why it affects high-frequency response.
- Be able to draw the equivalent capacitances seen at input/output.

6. Multistage Amplifiers

- Analyze a two-stage CE-CE amplifier.
- Analyze a CS-CS amplifier.
- Calculate overall voltage gain from individual stage gains.
- Understand loading between stages.

7. BJT Small-Signal / Hybrid- π Model

- Draw the hybrid- π model.
- Identify r_{π} , g_m , r_o and their roles.
- Understand transition frequency and high-frequency behavior.

8. Other Topics to Revise

- **Class-A Power Amplifier:** output power, efficiency and maximum-efficiency derivation.
- **Operational Amplifier Basics:** ideal characteristics, input bias current, offset voltage and frequency response.

■ 30-MINUTE LAST-MINUTE PLAN

Time	What to revise
0–7 min	CE amplifier numerical: midband gain + lower cutoff frequency
7–12 min	BJT frequency-response curve + f_L , f_H , bandwidth
12–16 min	MOSFET high-frequency model + upper cutoff
16–20 min	Current mirror + Widlar current source
20–24 min	Miller effect
24–27 min	Multistage amplifier
27–30 min	Hybrid- π model + Class-A / Op-Amp quick review

■ Final Checklist

- I can draw the BJT frequency-response curve.
- I can solve a CE amplifier gain/cutoff-frequency numerical.
- I can draw the hybrid- π model.
- I can explain current mirror and Widlar source.
- I can explain Miller effect.
- I can analyze a multistage amplifier.
- I know the key Class-A and Op-Amp basics.

Note: This is a PYQ/syllabus-based revision priority list, not a guarantee of the exact questions in the exam.